Long-Term Performance and Stability of Commercial Membranes during Produced Water Treatment by Air Gap Membrane Distillation

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Abstract

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Keywords

Air Gap Membrane Distillation; Produced water treatment; Membrane wetting; BTEX compounds; Membrane selection; Liquid Entry Pressure (LEP); Permeate flux; Salt rejection; Energy efficiency.

Introduction

Oil and natural gas extraction generates produced water as its main byproduct, with an estimated production of approximately 250 million barrels per day (39.75 million cubic meters) [1, 2]. In 2018, the global annual production of produced water was estimated at 6.8 billion cubic meters, a volume that tends to increase as oil fields mature, with water accounting for up to 98% of the fluids produced from mature reservoirs [3, 4]. In this context, the management of this effluent represents a significant technical and environmental challenge for the oil and gas industry.

Historically treated as a waste stream from the petroleum industry, produced water has increasingly been regarded as a resource with potential for reuse, particularly in reinjection operations, where it can contribute to residual oil mobilization and reservoir pressure maintenance while reducing freshwater consumption. Thus, produced water reuse can contribute to compliance with environmental regulations [5–7] while improving the efficiency of oil and gas production operations.

However, the feasibility of reinjection is directly associated with the effectiveness of produced water treatment. Produced water consists of a mixture of formation water, which is naturally present in petroleum reservoirs, and water used during production operations. Consequently, its composition varies depending on reservoir location, depth, age, and operational practices and may contain different concentrations of dispersed oil, dissolved salts, heavy metals, suspended solids, volatile organic compounds, naturally occurring radioactive materials, and production chemicals such as corrosion inhibitors and antiscalants [4, 8, 9]. High concentrations of these constituents can adversely affect the reservoir and production equipment [10, 11], making appropriate produced water treatment essential for its reuse.

Different treatment stages are employed in the petroleum industry, including primary, secondary, tertiary, and polishing processes [12–15]. Among the emerging technologies for produced water treatment and reuse, membrane distillation (MD) offers desirable characteristics, including high salt rejection, the ability to operate with highly saline solutions, and the potential for integration with low-temperature heat sources, enabling the utilization of waste heat [16–18].

Membrane distillation is a thermally driven separation process in which a hydrophobic microporous membrane separates two media maintained at different temperatures. The temperature gradient across the membrane establishes a partial vapor pressure difference, which provides the driving force for the separation process. MD can be operated in different module configurations, which differ primarily in the manner in which water vapor is condensed.

In air gap membrane distillation (AGMD), water vapor permeates through the membrane, crosses an air gap, and condenses on a cooled surface in contact with the cold side of the module, thereby preventing direct contact between the condensed permeate and the cooling fluid. The module configuration directly influences the heat and mass transfer mechanisms involved in the process [16–19].

Despite the favorable characteristics of membrane distillation for treating highly saline solutions, its application to produced water requires particular attention because interactions between the membrane and contaminants can compromise process performance through fouling, scaling, and wetting. These phenomena are influenced by both the morphological and surface characteristics of the membrane and the composition of the feed water [20–22].

Long-term studies have demonstrated stable MD operation with saline solutions, whereas investigations using produced water have reported flux decline, fouling, scaling, and membrane wetting associated with the complexity of the feed composition and pretreatment conditions [23–27]. However, among the studies analyzed, no systematic long-term assessment was identified that compares different commercial membranes in AGMD for produced water treatment and relates their morphological and surface characteristics to the temporal evolution of process performance and operational stability.

In this context, this study aims to investigate the performance and stability of five commercial membranes applied to produced water treatment by AGMD under long-term operation. The influence of operating time on permeate flux and conductivity is evaluated, together with the relationship between the morphological and surface characteristics of the membranes and their behavior during prolonged operation. The comparative analysis of the different membranes provides insight into their susceptibility to phenomena such as fouling, scaling, and wetting and seeks to identify membrane properties associated with enhanced process stability during produced water treatment.

Literature Review

Insert Literature Review here.

Methodology

Insert Methodology here.

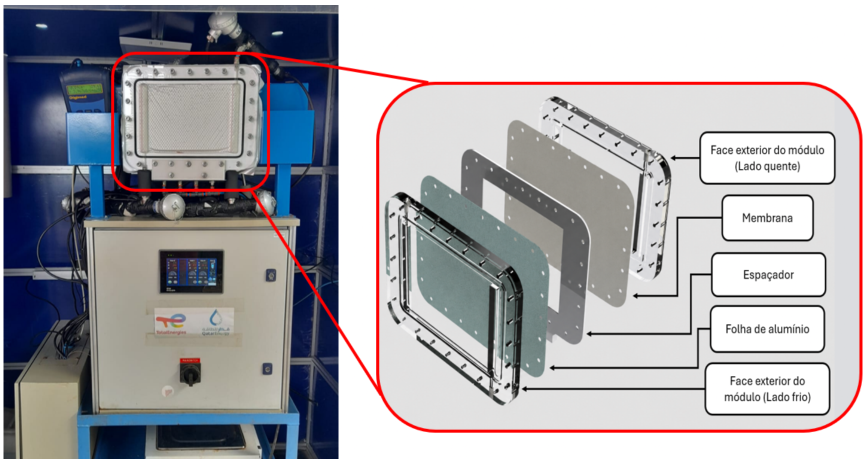

Figure 1: Módulo montado em bancada.

Results

Insert Results here.

Discussion

Insert Discussion here.

Conclusion

Insert Conclusion here.

Acknowledgements

Insert acknowledgements here.

Supporting Information

Insert supporting information description here.

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